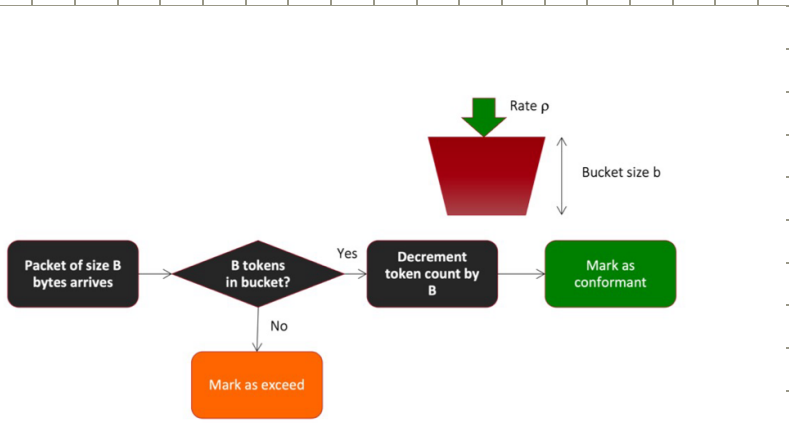
20/10/2026

S = token gen. rate

b = bucket size

(1 token = 1 Byte)



Pros and Cons

- One bucket refilled at fixed rate (CIR) → easy to implement
- Allows occasional bursts to the bucket size
- Packets simply marked as "non-conformant" → no possibility to differentiate packets
- Control only average rate and burst size
- Doesn't limit instantaneous peaks

Better model: 2R3C

Two-rate three colors

Params:
- CIR
- PIR: peak info rate
- CBS
- PBS: Peak Bucket size

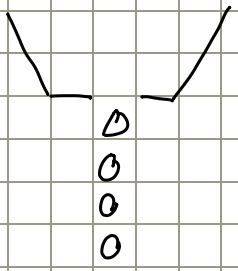
Suppose source rate at λ

- if $\lambda \leq$ CIR → all packets tagged as conformant
- if $\lambda >$ CIR but $\lambda \leq$ PIR → CBS empties, PBS does not
- $\lambda >$ PIR → both bucket emptying

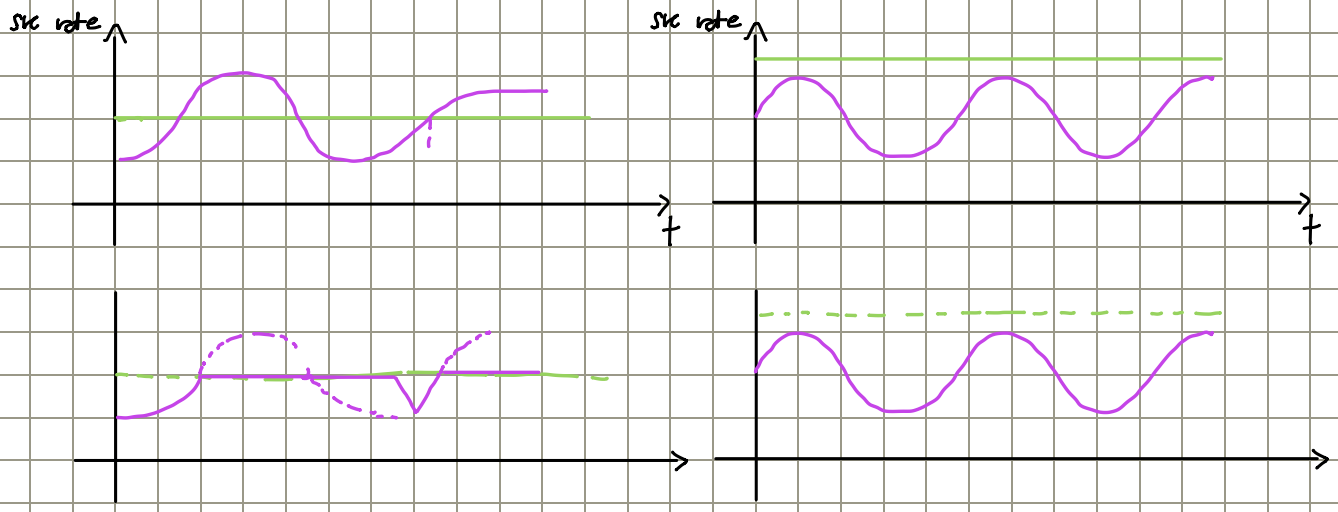
Token bucket for Traffic shaping

Leaky bucket

Token bucket filter → as a shaper



More used as a buffer

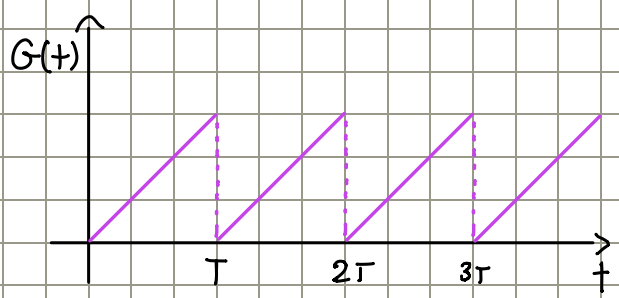


Example

LBF with $\rho = 1$ Mbps

$b = 1$ Kbyte

Source VBR $G(t) = R \cdot \mathrm{mod}(t, T)$



$$\mathrm{mod}(t,T) \triangleq \left(\frac{t}{T} - \left\lfloor \frac{t}{T} \right\rfloor\right) \cdot T \in [0,T)$$

$\lfloor \frac{t}{T} \rfloor$ → floor $\left(\frac{t}{T}\right)$

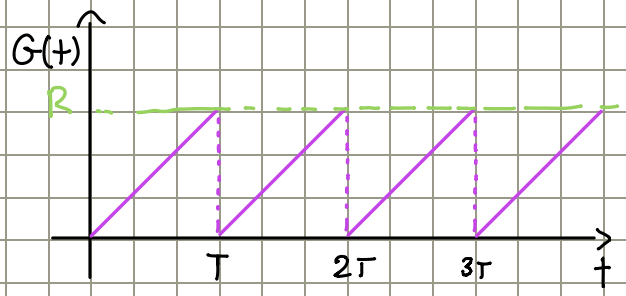
Given this input, what's the output?

Per period:

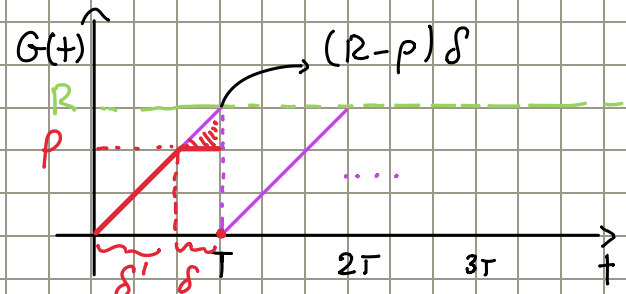
$B = RT$

Bucket can store at most $b = 1$ Kbyte $= 8000$ bits

If $R \leq \rho$ → filter transmit incoming traffic as it arrives
so output $= R$

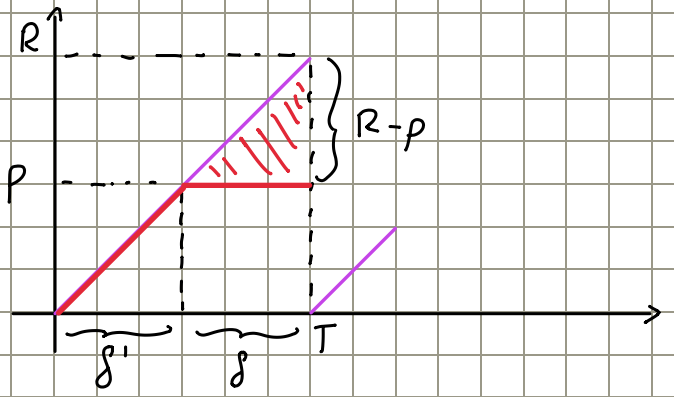


if $R > \rho$ → bucket accumulate traffic while transmitting at rate ρ



Buffer at time $t = T$

$$b(T) = \frac{\delta(R-\rho)}{2} = \frac{(R-\rho)^2}{2R} \cdot T$$



Assumption: empty bucket
If bucket was-empty:
higher rate.

$$b(t) \overset{\uparrow}{=} b(T+\varepsilon) = b(t) + \underbrace{\left(\frac{\varepsilon}{T} \cdot R - \rho\right)\varepsilon}_{G(T+\varepsilon)\cdot\varepsilon = G(\varepsilon)\cdot\varepsilon}$$

$t > T$

When $\frac{\varepsilon R}{T} \overset{?}{>} \rho \longrightarrow \varepsilon > \frac{\rho T}{R} < T$ before ending period, ... (finish with prof's transcription)

$$b(T+\varepsilon) = b(T) + \left(\frac{\varepsilon}{T} R - \rho\right)\varepsilon$$

$\varepsilon = ?$ s.t. $b(T+\varepsilon) = 0$

$$\varepsilon ? = \frac{R}{T} - \varepsilon\rho + b(T) = 0$$

$$\varepsilon^2 - \varepsilon\rho\frac{T}{R} + b(T)\cdot\frac{T}{R} = 0$$

$$\varepsilon_{1,2} = \frac{+\rho\cdot\frac{T}{R} \pm \sqrt{S^2\cdot\frac{T^2}{R^2} - 4b(T)\cdot\frac{T}{R}}}{2}$$

$$b(T) = \frac{(R-\rho)^2}{R}\cdot T$$

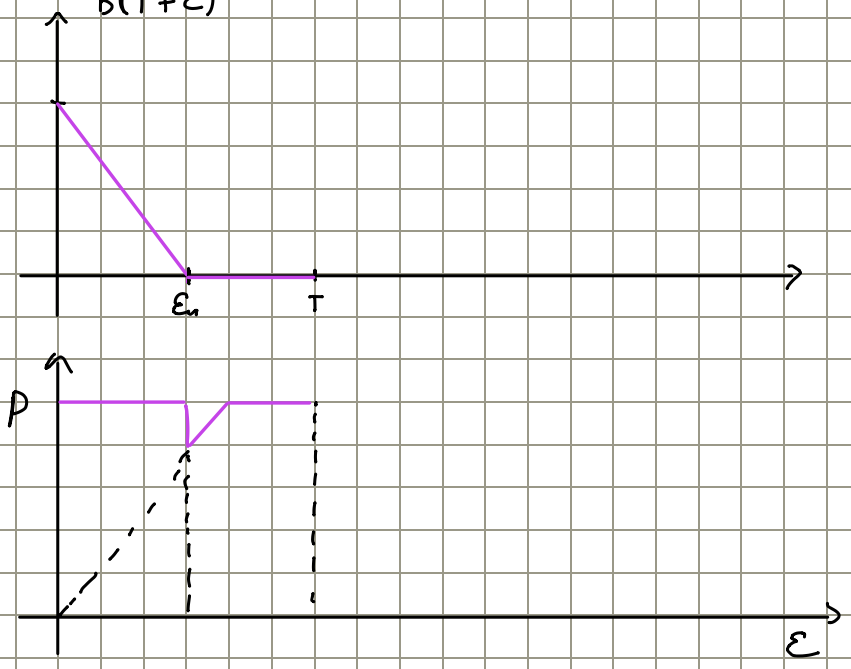
$$\varepsilon_{1,2} = \frac{\rho\cdot\frac{T}{R} \pm \sqrt{\left(\frac{\rho T}{R}\right) - 4\left(\frac{R-\rho}{R}\cdot T\right)^2}}{2}$$

If no solution → buffer no empty in that period

$\Delta < 0 \rightarrow \nexists\, \varepsilon : b(T+\varepsilon) = 0 \quad \Rightarrow G_{out}(t) \underset{t\to\infty}{\longrightarrow} \rho$

$$b(t) \underset{t\to\infty}{\longrightarrow} b$$

$\exists\, \varepsilon < T$



Check if exercise is meaningful and correct

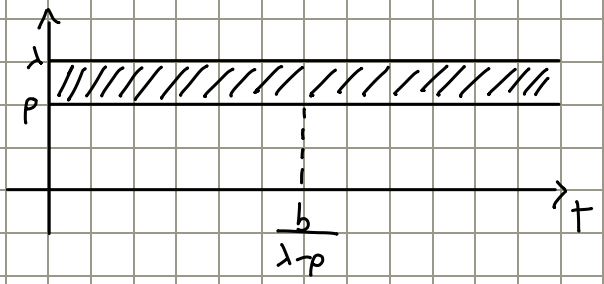
Leaky bucket vs Token bucket

LB: ρ

Source: CBR: $\lambda = 1,5\,\rho$

After a time $\frac{b}{\lambda - \rho}$ the bucket is full

the output rate is limited to ρ and the excess traffic $\lambda_d = 0,5\,\rho$ is dropped



Traffic Shaping

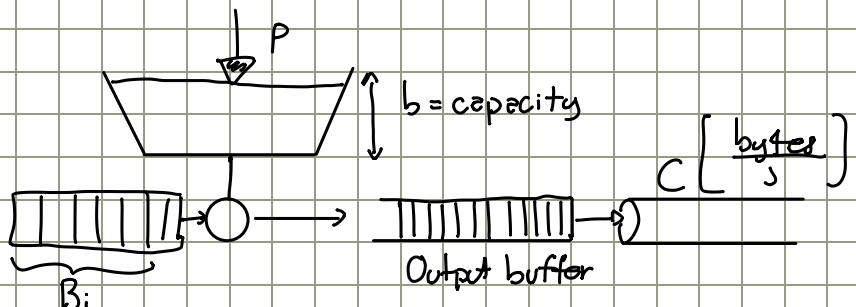
λ_a: average rate, long term average bit-rate

λ_p: PIR $> \lambda_a$

T_b: maximum burst duration, ...

Grace period T_g: max time during which a source can generate traffic exceeding the shaping without causing packet loss

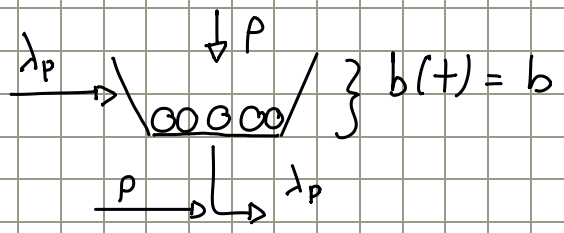
Max delay for packets: d_{max}, max delay for non dropped packets



Input buffer (link buffer)

1 token = 1 byte as general assumption

$\rho = \lambda_a$ (or $\geq$ at least)

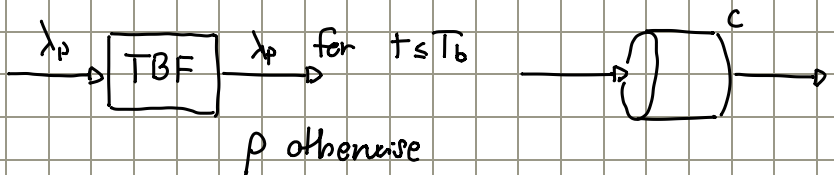


time to empty the bucket is $\frac{b}{\lambda_p - \rho} \geq T_b$

$b \geq T_b (\lambda_p - \rho)$

If $C > \lambda_p \rightarrow B_0 = 0$

$C \geq \lambda_a$
$C \geq \lambda_p$ output
$\lambda_a < C < \lambda_p \rightarrow$ buffer needed!



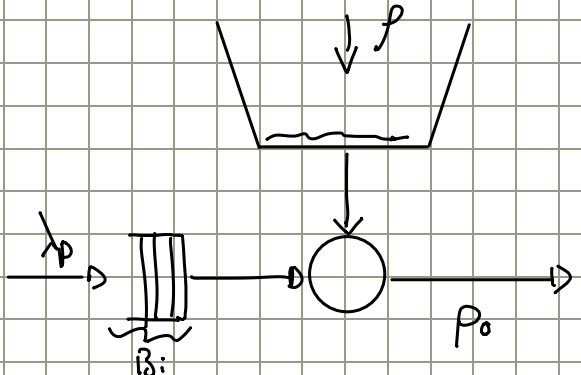
In time T_b the sauce generates $\lambda_p \cdot T_b$ bytes

The link transmits $C \cdot T_b$ bytes

$B_0 \geq (\lambda_p - C) \cdot T_b$

Note: we assume fluid model

We consider now the input buffer



$\lambda_p \cdot T_g \rightarrow$ generated

$\rho \cdot T_g \rightarrow$ forwarded

$B_i \geq (\lambda_p - \rho) T_g$

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